

BIO4AE3 - 14 Nov.

- move on experimental / obs. studies of host manipulation
- FISHER's theory of sex allocation.

↳ Sex ratio manipulation -

• FIELD EXPERIMENTS -

prevalence of behavioural change
correlations between infection
and behaviour
(change in host phenotype)

- PHYLOGENETIC studies -

given a host phylogeny, what is
the distribution of manipulation
across the phylogeny?

- MECHANISTIC studies.

Gammarus (copepod)

[modified escape responses
when infected]

similar behaviour by injecting serotonin.
BLOCK with octopamine

✱ Dicrocoelium dendriticum
(ant parasite)

TRANSCRIPTOMIC analysis...

infected ants in different behavioural
stages

PARASITIC CASTRATION

- reproductive organs are expendable to the parasite (don't impair viability)
 - divert energy into somatic growth.
 - GIGANTISM in castrated hosts.
- [• in one system, gigantism only occurs in local populations
Diplostomum phloxini

FISHER . sex ratio allocation.

✱ deviations from even sex ratios are interesting

↳ INBREEDING / closely related groups

(aphids, fig wasps)

↳ equal INVESTMENT into reprod of each sex.

↳ refers to sex ratio at end of parental investment, not at time of mating.

↳ CONDITION-dependent sex ratios.